

Durable Consumption during the Great Recession: the Role of Risk Heterogeneity

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Abstract

This paper studies the impact of risk heterogeneity on durable consumption over the business cycle. I examine household risk heterogeneity through the lens of labour contract type, and document that during the Great Recession, fixed-term contract workers reduced their car purchases significantly more than their permanent counterparts. Using an incomplete market model of durable and non-durable consumption with a dual labour market, I show that households that experienced an increase in downside income risk (such as permanent contract holders) adopted a "wait-and-see" strategy for their durable purchases, while households that experienced a decline in upside income risk (such as fixed-term contract holders) adopted a "wait-to-downgrade" approach. In addition, a car purchase subsidy has limited ability to stimulate durable consumption of households facing a decline in their upside income risk and is, *de facto*, partly transformed into a *downgrading subsidy*.

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1 Introduction

Durable consumption is one of the main driver of business cycle volatility. Consequently understanding how it interacts with employment and income risk is key to studying amplification mechanisms in recessions. What is more, households are heterogeneous in the level, nature, and fluctuations of their income risk. This is true, for example, of workers in different sectors of the economy or different cohorts.

This paper presents novel empirical evidence and a new theoretical framework for examining the impact of risk heterogeneity on durable consumption over the business cycle. I study households' risk heterogeneity through the lens of type of labour contracts. In many European countries including Italy, France, Spain, Sweden, Norway, Denmark and Finland, the labour market is segmented between workers employed with permanent contracts (i.e. indefinitely) and workers employed with fixed-term/temporary contracts (i.e. for a short and predetermined period only). Given their differing levels, nature and fluctuations of income and employment risk, comparing these workers is a promising approach to understanding the impact of risk heterogeneity. I start with some empirical motivation on durable and non-durable consumption between 2002 and 2014. I then build a dynamic heterogeneous agent model with durable and non-durable consumption where durable purchases are subject to a friction and the labour market is segmented into two types of contract.

Models studying durables over the business cycle focus on *ex-post* heterogeneity only (i.e. wealth heterogeneity resulting from transitory income shocks). Alternatively, in my framework, the type of contract represents a form of very persistent risk that lies between *full ex-ante* heterogeneity and the *ex-post only* heterogeneity of existing models. To the best of my knowledge, the impact of this more persistent heterogeneity has not been previously investigated. Furthermore, I analyse the extensive and intensive margins of durable consumption separately. The extensive margin of durable consumption corresponds to the share of households who purchase durable goods in a given period. The intensive margin is the amount spent on durable goods, conditional on purchasing them. I show that risk heterogeneity manifests differently across both adjustment margins.¹ This addresses a gap in the literature, which has mostly focused on the extensive margin of durable consumption.

There are three main findings. First, the increase in downside income risk is driving permanent contract workers' car response over the Great Recession. They adopt a "wait-and-see" strategy. Conversely, the decline in upside income risk is driving fixed-term contract workers' car response over the same period. They adopt

¹This is in line with Bertola, Guiso and Pistaferri (2005), who show that the two margins of durable consumption are driven by different economic factors.

a "wait-to-downgrade" approach. Second, around 40% of fixed-term contract workers' drop in car consumption is explained by a composition effect. The omission of this composition effect would result in a significant overestimation of the non-convex adjustment cost of cars. Third, a durable goods subsidy has limited ability to stimulate durable consumption among households experiencing a decline in upside income risk. The policy will, *de facto*, be partly transformed into a *downgrading subsidy*.

To conduct the empirical analysis I use data from the Survey of Households Income and Wealth (SHIW), conducted by the Bank of Italy. The SHIW dataset is notable for its comprehensive coverage of demographic and job-related characteristics, as well as detailed information on households' income, wealth, durable good expenditures, and non-durable consumption. Car purchases are used to measure households' durable consumption and I compare households whose members are employed with permanent contracts (referred to as permanent households for convenience) and households whose members are employed with fixed-term or temporary contracts (referred to as fixed-term households). The results highlight a stark and unevenly distributed drop in car purchases over the Great Recession. First, consistent with the findings of Attanasio et al. (2022), households sharply reduce their car purchases across both adjustment margins. Second, and this is a novel finding from this paper, the drop in car consumption is significantly larger for fixed-term households. During the Great Recession, the share of fixed-term households buying a car in a given year dropped by more than 40%, twice as much as for permanent households. On the intensive margin, fixed-term households reduced their spending by 17% compared to 12% for their permanent counterparts.

I then build and calibrate a model of household consumption and saving behaviour. The model is an incomplete market model in which households consume durable and non-durable goods. Durable purchases are subject to a non-convex adjustment cost *a la* Grossman and Laroque (1990). Following the empirical specification, I model two types of contract, each associated with a particular unemployment risk and income process. The policy functions derived from the model show that households update their durables according to a trigger-target (S,s) rule.² As durable investments are partially irreversible, households wish to limit the frequency of such purchases. This result is consistent with the empirical literature documenting lumpy durable consumption patterns.

I solve for the model's policy functions by implementing the NEGM+ algorithm

²(S,s) dynamics are studied by Attanasio (2000), Bar Ilan and Blinder (1992), Bertola et al. (2005), Caballero (1993), Caballero and Engel (1999), Eberly (1994) and Foote et al. (2000), among others.

developed in Druedahl (2021). This algorithm extends the endogenous grid-point method of Carroll (2006) to an economy with non-convexity and exploits the nested structure of the problem. The rich income and risk processes associated with each contract-type are estimated using the SHIW data. I then parameterize the model to be consistent with key features of the Italian economy between 2002 and 2006. Using the model, I simulate households' consumption response to a shock calibrated to replicate the Great Recession. Crucially, the model succeeds in matching the durable consumption patterns observed in the Great Recession along both adjustment margins for permanent and fixed-term households.

I then disentangle the impact of the change in households' risk from that of realised income losses in explaining the durable consumption contraction observed during the recession. To do this, I compare the baseline Great Recession experiment to a placebo in which households believe they are in a recession, but the underlying shocks hitting the economy are expansionary. The impact of risk is heterogeneous across contract types. For permanent households, change in risk explains half of the baseline decline in the extensive margin of car purchases but none of the decline in the intensive margin. The change in permanent households' risk profile during the Great Recession shifts down their trigger adjustment point (s), without impacting their optimal durables stock (S). For fixed-term households, change in risk accounts for about one-third of their baseline decline in cars extensive margin and 10% of their decline in cars intensive margin. Change in risk pushes down both their adjustment trigger (s) and their optimal durables stock (S). This heterogeneity is explained by the different nature of the risk faced by permanent and fixed-term contracts.

The increase in downside income risk is driving permanent households' car response over the Great Recession. The associated strategy is "wait-and-see". The income distribution of permanent households is negatively skewed. During the Great Recession, their probability of falling into unemployment or being downgraded to a fixed-term contract rises, increasing the variance and negative skewness of their income distribution. Since durables have non-convex adjustment costs, buying a car is a commitment that permanent households are less willing to make in an environment of high downside income risk. In such times, it is optimal for them to delay the purchase of durables and wait for uncertainty to subside.³ This "wait-and-see" strategy triggers a sharp but short-lived decline in the extensive margin of car purchases.

The decline in upside income risk is driving fixed-term households' car response over the Great Recession. The associated strategy is "wait to downgrade". Unlike their permanent counterparts, fixed-term households' income distribution is positively skewed. This is because, for them, the main source of large income changes is

³In line with Bernanke, (1983) and Dixit and Pindyck (1994).

being upgraded to a permanent contract. Fixed-term households have a lot of room to move up. However, during the Great Recession this upgrade probability falls significantly, leading to a 30% reduction in their income variance. Upside income risk has collapsed. As a result, households would like to reduce their stock of durable goods, but the non-convex adjustment cost makes selling cars extremely costly. As depreciation naturally reduces the value of cars over time, fixed-term households use this mechanism to downsize and save on transaction costs. In essence, they "wait-to-downgrade". Depreciation is a slow process, therefore fixed-term households' "wait-to-downgrade" strategy is persistent and continues throughout the recession.

Focusing on realised resource losses, I assess the importance of the composition effect. To be considered a permanent (fixed-term) worker, one must have been employed with a permanent (fixed-term) contract for at least 6 months in the year. Consequently, a proportion of households in the permanent (fixed-term) group will have experienced some spells of unemployment or fixed-term (permanent) contract during the period of interest. The number of households experiencing these alternative employment statuses depends on the aggregate state of the economy. Given the persistence of permanent contracts, the composition effect does not explain their car consumption patterns. Conversely, the composition effect accounts for around 40% of the decline in car purchases by fixed-term households along both adjustment margins. This is because, in recessions, these households are more likely to have experienced spells of unemployment and significantly less likely to access a permanent contract. If the researcher were to ignore the composition effect and tried to match the cars extensive margin of fixed-term households during the Great Recession, the non-convex adjustment costs would be estimated to be around 35%.

Finally, I use the model as a laboratory and examine the impact of a car subsidy equivalent to a payment of 5% of car expenditures in 2009. The results indicate the car subsidy merely induces households to front-load their car investments, leading to depressed car consumption in the post-program period. It also fails to stimulate the intensive margin of car consumption. Moreover, if the subsidy is targeted at households that have experienced a decline in upside income risk, it has even less ammunition to induce upward adjustment and may even be used by households to downsize their car. The policy is partly transformed into a *downgrading subsidy*.

This paper is related to the literature that studies lumpy durable consumption in macro models. Harmenberg and Oberg (2021) estimate the consumption response to an adverse labour market shock. Berger and Vavra (2015) show that non-convex adjustment frictions generate state-dependent responses to policy shocks, while McKay and Wieland (2021) highlight the inter-temporal trade-off for monetary policy in stabilising durables demand. Compared to these papers, my work focuses on risk

heterogeneity (by modelling a dual labour market) and separates the intensive and the extensive margins of durable consumption. In this respect, this paper is related to Attanasio et al. (2022), who are the first to study both adjustment margins in the context of a life-cycle model. They use the CEX to derive cohort and business cycle decomposition of durable and non-durable consumption profiles. More broadly, this work relates to the study of income and consumption dynamics (see, for example, Blundell and Preston (1998), Guvenen et al. (2014) and Guvenen et al. (2021)) and the impact of uncertainty on household and firm decisions (see Bayer et al. (2019), Bloom (2009), Bloom et al. (2018) or Storesletten, Telmer and Yaron (2004), among others).

The remainder of the paper is structured as follows. Section 2 presents motivational empirical evidence. Section 3 presents the model. Section 4 describes the calibration, numerical implementation and policy functions. The Great Recession experiment and the relative importance of the change in risk and the composition effect are presented in Section 5. Finally, Section 6 investigates the impact of a car subsidy policy. Section 7 concludes.

2 Motivating Evidence

To study the impact of risk heterogeneity over the business cycle, this paper focuses on labour contracts in Italy. The Italian labour market is segmented into those with permanent contracts and those with fixed-term/temporary contracts, who face different income and risk fluctuations.

I use data from the Survey on Households Income and Wealth (SHIW) conducted by the Bank of Italy. The main strength of the SHIW dataset is that, alongside many demographic and job related characteristics, it provides detailed information on households' income, wealth, durable goods expenditures and non-durable consumption. I use the SHIW waves between 2000 and 2014 to compare changes in employment status, income and consumption between workers employed with permanent contracts and those with fixed-term or temporary contracts. When variables are reported at the household level (e.g. consumption), I compare households whose members are employed with permanent contracts (called permanent households for convenience) and households whose members are employed with fixed-term or temporary contracts (called fixed-term households). The frequency of the data is biannual. A further presentation of the dataset, details on variables definition and sample selection, as well as summary statistics may be found in Appendix A.

Table 1 displays each group's changes in income, non-durable consumption and detailed car purchases over the Great Recession. The Great Recession refers to the

2008-2014 period (i.e. the 2008-2009 recession as well as the recessionary episode of 2011-2013 sometimes referred to as the sovereign debt crisis). One aim of this paper is to study the extensive and the intensive margin of durable consumption separately. Studying changes in the intensive margin on a basket of goods is misleading.⁴ Therefore, I restrict the durables to a single good: cars. I choose this particular item as - abstracting from houses as is done in this paper - cars represent the largest durable good purchased by households (between 2002 and 2014 cars alone accounted for roughly 30% of all durable purchases). In the remainder of the paper, I use cars and durables interchangeably.

	Cars ext. margin (€)		Cars int. margin (€)		Non-dur. cons. (€)		Income (€)	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	15.76	12.53	6,869	5,458	15,549	11,180	32,012	17,199
Recession	12.42	6.96	6,026	4,507	15,192	11,175	29,976	15,418
Change	-0.21	-0.44	-0.12	-0.17	-0.02	0.00	-0.06	-0.10

Notes: Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Boom is 2002-2006, Recession is 2008-2014. Non-dur. cons. stands for non-durable consumption, ext. for extensive and int. for intensive. Income is income from labour and transfers. Top and bottom 1% are winsorised. Households sampling weights are used.

Table 1: Consumption and Income Response to the Great Recession

Table 1 highlights a stark and unevenly distributed drop in car purchases over the Great Recession. First, in line with the findings of Attanasio et al. (2022), households strongly decreased their car purchases across both adjustment margins. Second, and this is a novel finding from this paper, the car consumption drop is significantly larger for fixed-term households. Over the Great Recession, the share of fixed-term households buying a car in a given year dropped by more than 40%, twice as much as for permanent households. On the intensive margin, fixed-term households reduced their expenses by 17% compared to 12% for their permanent counterparts. Finally, both groups more or less maintained their non-durable purchases over the recession. This holds despite a drop in mean income from labour and transfers of 6% and 10% for permanent and fixed-term households respectively.

Beyond income and consumption patterns, different types of contract also face heterogeneous levels and fluctuations of employment risk. Table 2 shows workers' employment status transition probabilities before and during the Great Recession.

⁴If agents buy a car in a given year and a sofa in the following year, it does not mean that they decreased their intensive margin of durable consumption. The drop in the value of the purchases is simply a reflection of different types of investment.

In the expansionary period, workers employed with fixed-term contracts had a 49% probability to be employed with a permanent contract in the next wave of the survey (i.e. two years later) and a 15% probability to become unemployed at the same time horizon. During the Great Recession, the two-year horizon probability to upgrade to a permanent contract dropped to 33%, while the unemployment risk increased to 20%. These workers record a significant drop in their upside employment risk and a rise in their downside employment risk. Permanent contract holders cannot upgrade to a more stable contract type, but they can either downgrade to a fixed-term contract or become unemployed. At a two-year time horizon, these events occurred with a probability of 4% and 3% in boom, and 6% and 5% in recession. As a result, the downside employment risk for permanent workers increased during the Great Recession.

	Up. to perm.		Down. to f.t.		Down. to u.	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	-	0.49	0.04	-	0.03	0.15
Recession	-	0.33	0.06	-	0.05	0.20

Notes: Perm. stands for permanent contract, F.t. for fixed-term or temporary contract, and u. for unemployed. Up. stands for upgrade and Down. for downgrade. Boom is 2002-2006 and Recession is 2008-2014.

Table 2: Probability to Change Employment Status (Two-year Horizon)

3 The Model

I formulate a model of household consumption and saving behaviour in which the labour market is segmented into two types of contracts, each associated with a particular unemployment risk and income process. The model is an incomplete market model where households consume durable and non-durable goods. Durable purchases are subject to a market friction. This specification is close to Berger and Vavra (2015) or Harmenberg and Oberg (2021).

3.1 The Household Problem

The economy is populated by a continuum of *ex-ante* identical households of measure one indexed by $i \in [0, 1]$. Households are infinitely lived, time is discrete and a period is a quarter. They supply labour inelastically.

3.1.1 Preferences

Households derive utility from their non-durable consumption (c_t) and their stock of durable goods (D_t). They discount the future at rate β . The value function of household i can be written as:

$$V_i = E_0 \max_{\{c_{it}, D_{it}\}} \sum_{t=0}^{\infty} \beta^t u(c_{it}, D_{it})$$

$$\text{with } u(c_{it}, D_{it}) = \frac{[c_{it}^\alpha D_{it}^{(1-\alpha)}]^{(1-\sigma)}}{1-\sigma}$$

where α is the weight of non-durable consumption in the utility function and σ is the coefficient of relative risk aversion.⁵

3.1.2 Idiosyncratic risk - employment risk

Households face an idiosyncratic employment risk. In a given period, a household can be either employed with a permanent contract, employed with a fixed-term contract or unemployed. Households with fixed-term contracts face a larger risk of becoming unemployed than households holding permanent contracts. Transitions between the three employment states follow a Markov process.

3.1.3 Idiosyncratic risk - income risk

Households also face a degree of labour income risk when they are employed. The logarithm of income follows an autoregressive process of order one given by:

$$\log(y_{it}) = \mu + \rho \log(y_{it-1}) + \xi_{it}$$

$$\text{with } \xi_{it} \sim \mathcal{N}(0, \sigma_\xi^2)$$

where μ , ρ and σ_ξ^2 are the intercept, persistence and variance of the household's income process.

When unemployed, households receive unemployment benefit ($y_{it} = ub$) with probability p_{ub} . Alternatively, they receive a minor subsistence allowance ($y_{it} = sub$) with probability $1 - p_{ub}$.

⁵The evidence in Piazzesi and Schneider (2007) and in Ogaki and Reinhart (1998) motivates the choice of the Cobb-Douglas utility function.

3.1.4 Aggregate risk - employment transition

In addition to households' idiosyncratic income and employment risk, the economy is either in good aggregate state (called boom) or in bad aggregate state (called recession). These aggregate states are characterised by two distinct matrices governing transitions between permanent employment, fixed-term employment and unemployment, as well as different levels of unemployment benefit.

3.1.5 Aggregate risk - wealth shocks

Asset markets are incomplete and households may self-insure against employment and income risk by saving in a composite asset a_t . The composite asset is made of securities valued at price q_t^s , real estate valued at price q_t^h , and some risk-free asset (deposit and valuables) valued at price 1.

Securities price is subject to some aggregate fluctuations following:

$$\log(q_t^s) = \rho \log(q_{t-1}^s) + \xi_{it}^s$$

$$\text{with } \xi_{it}^s \sim \mathcal{N}(0, \sigma_{\xi^s}^2)$$

where ρ and $\sigma_{\xi^s}^2$ are persistence and variance parameters.

Similarly, real estate price is subject to some aggregate fluctuations following:

$$\log(q_t^h) = \rho \log(q_{t-1}^h) + \xi_{it}^h$$

$$\text{with } \xi_{it}^h \sim \mathcal{N}(0, \sigma_{\xi^h}^2)$$

where ρ and $\sigma_{\xi^h}^2$ are persistence and variance parameters. I assume that securities and real estate price processes are independent.

Finally, the composite asset is bought and sold each period at price q_{jt}

$$q_{jt} = \omega_j^s * q_t^s + \omega_j^h * q_t^h + (1 - \omega_j^s - \omega_j^h) * 1$$

where ω_j^s and ω_j^h are the weights of securities and real estate in the households' portfolio. I assume that these weights are exogenous and depend on households' employment status j . Borrowing is not allowed in this economy.

3.1.6 Durable goods

Households may also self-insure by accumulating durable goods sold at price p . Durables stock D_t cannot be negative and depreciates at rate δ . Moreover, durable

purchases are subject to a friction. When households decide to adjust their stock of durables, they have to pay a non-convex adjustment cost τ , which is proportional to the stock of durable goods held by the household before adjusting. This adjustment cost follows the specification of Grossman and Laroque (1990) and ensures to reproduce the lumpy patterns of durable consumption documented in the microdata.

3.1.7 Budget constraints

The budget constraint of a household deciding not to adjust their stock of durables is:

$$a_{it}q_{jt} + c_{it} \leq (1 + r)q_{jt}a_{it-1} + y_{it}$$

where r is the interest rate on the composite asset.

Conversely, the budget constraint of a household deciding to adjust their stock of durables is:

$$a_{it}q_{jt} + c_{it} + pD_{it} \leq (1 + r)q_{jt}a_{it-1} + y_{it} + (1 - \tau)(1 - \delta)pD_{it-1}$$

3.2 Recursive Formulation of the Problem

V is the value function of a household. For concision, the i subscripts are dropped.

$$V(m, D, y, j; \Omega) = \max\{V^{keep}(m, D, y, j; \Omega), V^{adj}(x, y, j; \Omega)\}$$

$$s.t \quad x = m + (1 - \tau)(1 - \delta)pD$$

Where V^{keep} is the value function of a household who does not adjust durables and V^{adj} is the value function of a household who adjusts durables. m is the household's cash-in-hand and x is the cash-in-hand available to the household after having sold their beginning-of-period stock of durables D . y is income, j is the employment status of the household, and Ω is the vector of aggregate states of the economy (boom or recession for employment risk, price of securities q^s , and price of real estate q^h).

The keeper's problem is:

$$V^{keep}(m, D, y, j; \Omega) = \max_{a, c} U(c, D') + \beta E[V(m', D', y', j'; \Omega')]$$

$$s.t \quad qa = m - c$$

$$D' = (1 - \delta)D$$

$$m' = (1 + r)q'a + y'$$

$$\begin{aligned}
q &= \omega_j^s q^s + \omega_j^h q^h + (1 - \omega_j^s - \omega_j^h) \\
q' &= \omega_{j'}^s q^{s'} + \omega_{j'}^h q^{h'} + (1 - \omega_{j'}^s - \omega_{j'}^h) \\
y' &\sim \Upsilon(y) \\
j' &\sim \Phi(j) \\
\Omega' &\sim \Gamma(\Omega)
\end{aligned}$$

where Υ is the conditional distribution of the idiosyncratic labour income, Φ is the conditional distribution of the employment status, and Γ is the conditional distribution of the aggregate states.

The adjuster's problem is:

$$\begin{aligned}
V^{adj}(x, y, j; \Omega) &= \max_{a, c, D'} U(c, D') + \beta E[V(m', D', y', j'; \Omega')] \\
s.t \quad qa &= x - c - pD' \\
m' &= (1 + r)q'a + y' \\
q &= \omega_j^s q^s + \omega_j^h q^h + (1 - \omega_j^s - \omega_j^h) \\
q' &= \omega_{j'}^s q^{s'} + \omega_{j'}^h q^{h'} + (1 - \omega_{j'}^s - \omega_{j'}^h) \\
y' &\sim \Upsilon(y) \\
j' &\sim \Phi(j) \\
\Omega' &\sim \Gamma(\Omega)
\end{aligned}$$

Following the nested structure in Druedhal (2021), the adjuster's problem can be viewed as a sequential problem. The household first chooses how much durable goods to buy or sell, and then chooses non-durable consumption. I rewrite the adjuster's problem as:

$$\begin{aligned}
V^{adj}(x, y, j; \Omega) &= \max_{D'} V^{keep}(m, D, y, j; \Omega) \\
s.t \quad D' &= (1 - \delta)D \\
m &= x - pD'
\end{aligned}$$

4 Parameterization, Numerical Implementation and Policy Functions

4.1 Parameterization

I parameterize the model to be consistent with key features of the Italian economy before the Great Recession (2002-2006). One period in the model is a quarter. I use a mixed parameterization strategy. A subset of parameters is fixed using standard values and the literature. Another set of parameters is calibrated to match moments from the Italian economy outside the model. The remaining parameters are jointly calibrated using the method of simulated moments inside the model. The parameter values are summarized in Table 3. Table 4 shows the targeted moments.

4.1.1 Calibration of the employment risk

Quarterly employment risk is externally calibrated to match bi-annual transitions from the SHIW data. As mentioned above, households' transitions between the three employment states - employed with a permanent contract, employed with a fixed-term contract, and unemployed - are governed by two Markov processes: one in boom and one in recession. As type of contract is only available from 2000 in the data, I do not have access to an appropriate recession period before the Great Recession. Consequently, I use the SHIW data before 2008 to calibrate the model's quarterly transitions in boom, and the SHIW data between 2008 and 2014 to recover the model's transitions in recession. I use the method of matching simulated moments.⁶ Appendix B shows the bi-annual employment transitions from the data and the model. The quarterly transition matrices obtained using this procedure are reported below. Rows represent employment state today and columns employment state next period. p stands for employed with a permanent contract, $f.t$ employed with a fixed-term contract and u unemployed.

$$\mathbf{P}_{\text{boom}} = \begin{matrix} & \begin{matrix} p & f.t & u \end{matrix} \\ \begin{matrix} p \\ f.t \\ u \end{matrix} & \begin{pmatrix} 0.988 & 0.008 & 0.004 \\ 0.104 & 0.858 & 0.038 \\ 0.029 & 0.042 & 0.929 \end{pmatrix} \end{matrix}, \quad \mathbf{P}_{\text{recession}} = \begin{matrix} & \begin{matrix} p & f.t & u \end{matrix} \\ \begin{matrix} p \\ f.t \\ u \end{matrix} & \begin{pmatrix} 0.984 & 0.010 & 0.006 \\ 0.063 & 0.894 & 0.042 \\ 0.019 & 0.030 & 0.951 \end{pmatrix} \end{matrix}$$

⁶I use the diagonal matrix of inverses of the moments' relative variances as a weighting matrix. Moments' variances are obtained with a thousand-repetition bootstrapping procedure.

In a given aggregate state, permanent contract holders face a significantly lower risk of losing their job than fixed-term contract workers. Unemployed workers have higher chances to find fixed-term contracts than permanent ones to exit unemployment. Comparing aggregate states, employed workers (with any type of contract) have higher risk of falling into unemployment in recession. Similarly, it is harder for unemployed workers to transition back into employment in recession than in boom. Finally, moving from a fixed-term contract to permanent employment is significantly less likely when the economy is in recession.

4.1.2 Calibration of the income processes

Income processes are calibrated using the SHIW data between 2000 and 2006. In period t , the logarithm of household i 's income $\log(y_{it})$ is given by:

$$\begin{aligned}\log(y_{it}) &= Z'_{it}\beta + \tilde{y}_{it} \\ \tilde{y}_{it} &= P_{it} + \epsilon_{it} \\ P_{it} &= \tilde{\mu} + \tilde{\rho}P_{it-1} + u_{it} \\ \epsilon_{it} &\sim i.i.d, \quad u_{it} \sim \mathcal{N}(0, \sigma_u^2)\end{aligned}$$

where Z_{it} is a set of household's observable characteristics including type of contract and other demographic variables. Income residual $\tilde{y}_{it}$ has a persistent component P_{it} which follows an auto-regressive process of order one, and some i.i.d measurement error ϵ_{it} . Income residuals are obtained by performing a standard OLS regression of the logarithm of individuals' labour income on year dummies, type of contract, gender, age, age squared, education, region, and size of city. I then use variance covariance identifying restrictions to recover the persistent component's intercept, persistence and variance parameters. After discarding measurement error, the income process has a quarterly persistence a little above 0.98 and a variance of 0.0034. These estimates are in line with the boom quarterly values of Storesletten et al. (2004). Ultimately, type of contract specific average fitted values from the regression are added to the residuals to obtain the final grids of income for permanent and fixed-term workers. Values are normalised by the average quarterly income of permanent contract workers. Additional details are reported in Appendix B.

Unemployment benefit levels are set to reproduce mean unemployment transfers of households who do receive unemployment benefit. For boom, I take the SHIW data between 2002 and 2006. For recession, I consider observations between 2008 and 2014. The probability of receiving unemployment benefit is chosen to match the period's SHIW unemployment benefit coverage rate of roughly 12%.

4.1.3 Calibration of the wealth shocks

Securities price fluctuations are calibrated using the log of S&P500 adjusted close prices deflated by CPI between January 1985 and December 2007. After removing a linear trend, I use a standard OLS regression to recover the AR(1) persistence parameter and the variance of the residuals. A similar procedure is applied for real estate price fluctuations using the log of the real residential property price index for Italy between January 1980 and December 2007. Weights of securities and real estate in households' portfolio come from the SHIW data between 2002 and 2006. Permanent households' average portfolio consists of 5% of securities (including government bonds), 59% of real estate, and 36% of deposits and other real assets. Households do not hold any business assets as I excluded self-employed from the sample. Fixed-term households' portfolio consists of only 1% of securities (including government bonds), 46% of real estate, and 52% of deposits and other real assets. Unemployed households' portfolio weights are assumed to be the same as fixed-term households'.

4.1.4 Other parameters fixed outside the model

I set the coefficient of relative risk aversion to 2. The interest rate on the composite asset is set to .01, which delivers an annual interest rate of approximately 4%. Price of durable goods is normalised to 1. In line with Harmenberg and Oberg (2021) and Attanasio et. al. (2022), I set car net depreciation to 10% per year. The subsistence allowance given to households who do not receive unemployment benefit is set to 0.06 (corresponding to €100 per month). Finally, I set the transitions between booms and recessions to match the average length of recessions (7.5 periods) and the share of total time spent in recessions (43%) in Italy between 1960 and 2016.⁷ I use the OECD based recession indicators for Italy computed by the Fed of Saint Louis. All the calibrated parameters are summarised in Table 3.

4.1.5 Parameters jointly calibrated inside the model

I jointly calibrate the remaining parameters inside the model using the method of simulated moments on the SHIW data between 2002 and 2006.⁸ Because of the non-convex adjustment cost, the model's wealth and durables stock are strongly path dependent. Consequently, extra care should be taken when setting up the

⁷Following the method in Krueger, Mitman and Perri (2016).

⁸As for the income process, I use a bootstrapping procedure with a thousand repetitions to get the moments' variances. I then take the diagonal matrix of inverses of the moments' relative variances as a weighting matrix.

Parameter	Value	Description	Target
Households			
β	0.979	Discount factor	See Table 4
σ	2.00	Relative risk aversion	Standard value
r	0.01	Interest rate	Annual interest rate of 4%
α	0.934	Weight of n.d.c. in utility	See Table 4
τ	0.065	Dur. adjustment cost	See Table 4
δ	0.025	Depreciation rate	Annual depreciation of 10%
p	1.00	Dur. price	Normalisation
ub_{boom}	0.32	U.b in boom	Mean u.b 2002-2006
$ub_{recession}$	0.25	U.b in recession	Mean u.b 2008-2014
sub	0.06	Subsistence allowance	€100 per month
p_{ub}	0.12	Probability to get u.b	u.b coverage rate 2002-2014
Agg. state			
ρ_{bb}	0.90	Boom to boom transition	Time spent in rec.
ρ_{rr}	0.87	Rec. to rec. transition	Average length of rec.

Notes: All values are reported at the quarterly frequency of the model. N.d.c. stands for non-durable consumption, Dur. stands for durables, u.b stands for unemployment benefit, Rec. stands for recession.

Table 3: Parameters

initial condition in the simulations. Starting from the stationary distribution for durables and wealth, I simulate the model for the 1980-2001 period feeding in the path of realised aggregate shocks for Italy (OECD based recession indicators). I then compute the model’s moments in the expansion period between 2002 and 2006. This procedure ensures that the moments from the model are comparable with their data counterparts. The jointly calibrated parameters are households’ discount factor (β), cars non-convex adjustment cost (τ), and the weight of non-durables in the utility function (α). The set of targets and the associated estimates are displayed in Tables 3 and 4. The calibration results are well in line with estimates from the literature.

Target	Model	Data	Source
Share of hh. buying a car in a year	0.16	0.16	SHIW 02-06
Mean car expenses norm. by income	0.33	0.33	SHIW 02-06
Median wealth norm. by income	3.55	3.55	SHIW 02-06

Notes: Hh. stands for households and Norm. stands for normalised. Income refers to yearly income from labour and transfers. The mean of car expenses is conditional on buying a car.

Table 4: Targeted Moments

4.2 Model Fit: Non-targeted Moments

This subsection presents how the model fits some important moments that were not explicitly targeted in the calibration. Table 5 displays these cross-sectional moments in the model, and in the data. The data corresponds to years between 2002 and 2006. The model simulates the same time period, with the procedure to set up the initial condition explained in the previous subsection.

As shown on the top half of Table 5, the model is particularly successful in reproducing non targeted moments by type of contract. This is key as the aim of the paper is to analyse and compare permanent and fixed-term households’ consumption patterns over the Great Recession. The model reproduces the share of permanent and fixed-term households who adjust their car, as well as ratios of car purchases, income and consumption by type of contract. The detailed calibration of contract specific income and employment risk and of cars adjustment cost explains why these non-targeted income and consumption moments are well aligned with the data. As is common in this type of models, I do not capture the high degree of wealth concentration among the very rich. Consequently the model underestimates permanent

households' wealth relative to that of fixed-term households.

The bottom part of Table 5 displays moments for the entire population. The model matches each of these non-targeted moments closely with the exception of car purchases' cross-sectional standard deviation. This was expected as, in the solution method, households choose car purchases on grid-points. The variance of such expenses is therefore underestimated.

Moment	Model	Data
By type of contract		
Share of car adjusters - perm.	0.17	0.16
Share of car adjusters - f.t.	0.13	0.13
Ratio perm./f.t. - car purchases	1.27	1.26
Ratio perm./f.t. - income	1.86	1.86
Ratio perm./f.t. - cons.	1.39	1.40
Ratio perm./f.t. - wealth	1.31	2.00
Overall population		
Mean ratio car purchases / cons.	0.32	0.43
Mean ratio car purchases / stock cars	0.73	0.73
Mean ratio stock cars / cons.	0.42	0.47
Mean share of cars in total assets	0.11	0.07
Share of car downgraders	0.005	0.006
Sd. of income	0.47	0.55
Sd. of car purchases	0.32	0.78
Sd. of cons.	0.34	0.45

Notes: Cons. stands for non-durable consumption. Ratio perm./f.t. refers to the mean of the variable for permanent households over the mean of the variable for fixed-term households. Sd. is the cross sectional standard deviation normalised by the mean. Income refers to income from labour and transfers. Car purchases are conditional on buying a car. Total assets refer to wealth + stock of cars.

Table 5: Non-targeted Moments

4.3 Numerical Implementation

I solve for the model's policy functions in partial equilibrium by implementing the NEGM+ algorithm developed in Druedahl (2021). This algorithm extends the en-

ogenous grid-point method of Carroll (2006) to an economy with non-convexity and exploits the nested structure of the problem. An additional layer of optimisation is attained with an enhanced interpolation method. I solve for households' policies on a 30-point grid for durables stock, and 200-point grids for regular cash-in-hand, cash-in-hand after reselling the stock of durables, and liquid assets. As there are three grids to represent the same dimension (cash-in-hand, cash-in-hand after reselling the stock of durables, and liquid assets), particular care should be taken in setting up these three grids relatively to each other. The top of the grid for cash-in-hand after selling the stock of durables should be larger than that of the assets grid. This is the case as households should use some of their resources for consumption. Moreover, setting the maximum level of cash-in-hand after selling durables equal to the sum of the tops of the cash-in-hand and durables stock grids ensures that the grid is wide enough for comparing the adjust and keep cases. The labour income autoregressive processes for permanent and fixed-term workers are each discretized into five states Markov processes using Rouwenhorst's method. In addition to the ten employment states, there also exists two unemployment states (unemployed households who receive unemployment benefit or not) bringing the number of idiosyncratic states to 12. The economy can be in boom or in recession (high or low employment risk). Additionally there are 9 aggregate states for wealth shocks as securities and real estate price processes are each discretized into three states Markov processes with Rouwenhorst method. I iterate the value function until convergence using the absolute value of the largest difference as an error metric and a tolerance level of $1e-3$.

4.4 Decision Rules

Figure 1 plots choices of durables stock as a function of durables stock at the start of the period for permanent (in blue) and fixed-term households (in yellow).⁹ These policy functions illustrate that households update their durables following trigger-target (S,s) rules. As durable investments are subject to a non-convex transaction cost, agents wish to limit the frequency of such adjustments. Households will decide on a minimum stock of durables under which they do not want to sink and a maximum stock of durables above which it is inefficient for them to go. As long as their durables stock is between these two bonds, they will not make any adjustment and simply let their durables depreciate. Once the stock of durables depreciated below the lower trigger point, households become willing to pay the adjustment cost to bring their durables up to a target value. Figure 1 shows adjustment trigger and

⁹The other states are held fixed to mean cash-in-hand, mean income, boom aggregate state for employment risk, and median prices for securities and real estate.

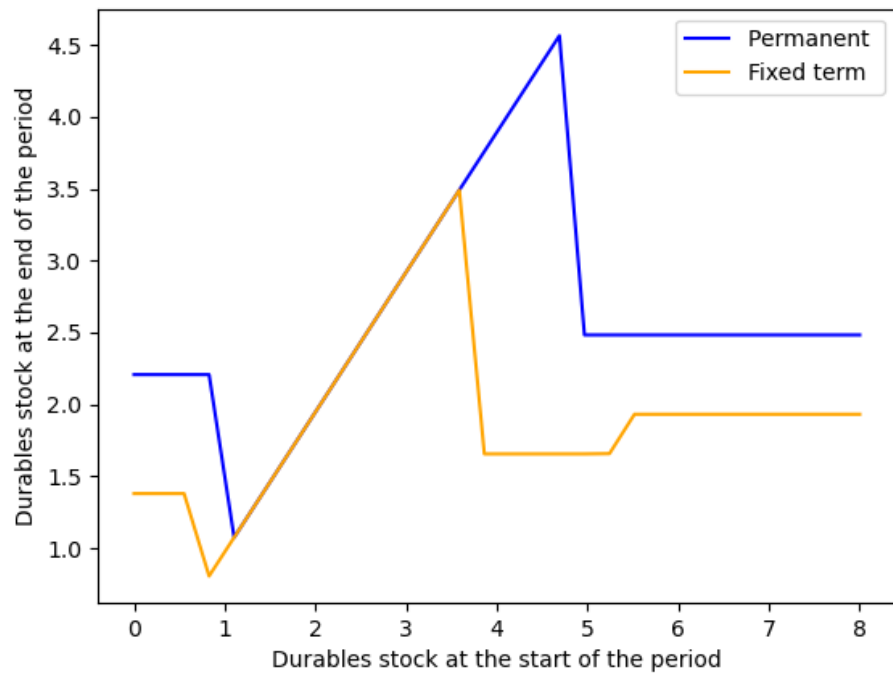
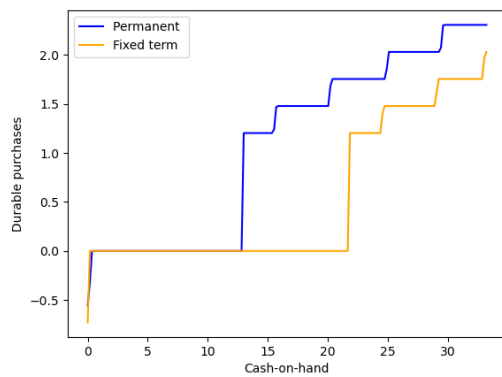
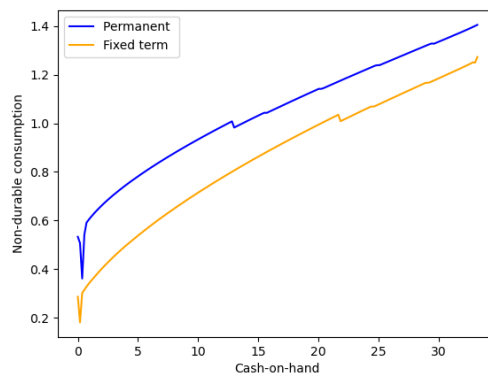


Figure 1: Policy Function: Stock of Durables



(a) Durable Purchases



(b) Non-durable Consumption

Figure 2: Policy Functions

target points for buying and selling durables. In between these points, households are in the inaction region.

Figure 1 also highlights the difference between permanent and fixed-term households. Permanent households' optimal stock of durables is higher than that of fixed-term households. Moreover, fixed-term households will wait longer before buying new durables and will sell their existing stock faster than their permanent counterparts. Fixed-term households' inaction region is shifted downwards. The discrepancy between the two types of contract's durable consumption is also illustrated in Panel a of Figure 2 where durable purchases are plotted against cash-in-hand.¹⁰ Here, the inaction region of fixed-term households is larger than that of the permanent group as they will need a higher level of cash-in-hand to prompt an upward adjustment. Moreover, conditional on buying durables, the value of fixed-term households' purchases is lower than permanent households' tickets. For very low levels of cash-in-hand, durable purchases are negative. This is a region where households sell their current durables to afford non-durable consumption.

Finally, Panel b of Figure 2 plots non-durable consumption as a function of cash-in-hand.¹¹ As non-durable consumption isn't subject to frictions, there are no inaction region or (S,s) type behaviours. The drop in consumption at the start of the x-axis corresponds to when households stop selling their existing durables to afford non-durable consumption. Similarly the downward step in consumption at higher levels of cash-in-hand indicates that some resources are shifted towards buying durable goods.

5 The Great Recession

5.1 The Great Recession - Baseline

I simulate the model's response to a shock calibrated to reproduce the Great Recession and study the durable and non-durable consumption patterns of permanent and fixed-term households. As in the calibration procedure, I take care when setting up the simulations' initial condition because durable purchases are path dependent. I start the simulations by feeding in realised booms and recessions between 1980 and 2001 using Italy's OECD based recession indicators. I then simulate a boom for the 2002 to 2007 period where securities price is held median and real estate price is high

¹⁰The other states are held fixed to mean stock of durables, mean income, boom aggregate state for employment risk, and median prices for securities and real estate.

¹¹Once again the other states are held fixed to mean stock of durables, mean income, boom aggregate state for employment risk, and median prices for securities and real estate.

(to be consistent with the surge in real estate’s value prior to the Great Recession). I then simulate the Great Recession from 2008 to 2014 with recession specific employment risk, low securities and real estate prices, as well as an additional MIT shock to labour income calibrated to reproduce the labour income drop experienced by fixed-term and permanent households during this period. The length of the episode and the extra income drop confer an exceptional nature to the Great Recession.

The aim of this exercise is to evaluate if the model is able to reproduce the empirical facts in the motivation section of the paper. Therefore, the data sample selection should be reproduced in the model simulations. In the data, households report their employment status for the majority of the year. Consequently, households who belong to the fixed-term (permanent) group could have been employed with a permanent (fixed-term) contract or unemployed for a small time during the year. The same is true for the model simulations.

The results of the Great Recession experiment are reported in the first panel of Table 7 where the estimates from the motivation section (Table 1) are compared to their model counterparts. Each type of household’s drop in labour income are targeted moments, while the changes in car and non-durable consumption are not targeted. The model is successful in reproducing the patterns observed during the Great Recession. Over the period, the model share of households purchasing a car dropped by 22% and 42% for permanent and fixed-term households, against 21% and 44% in the data. Moreover, fixed-term household’s intensive margin of car purchases decreased by 19% in the model, compared to 17% in the data. These figures are 9% and 12% for permanent contract holders. Finally the model closely tracks the change in permanent households’ non-durable consumption, while overestimating a little the contraction for fixed-term households.

5.2 The Role of Risk

The previous subsection established that the model is successful in reproducing the observed consumption patterns of permanent and fixed-term households over the Great Recession. I now want to disentangle the impact of the change in households’ risk from that of realised income losses. To this end, I compare my baseline Great Recession experiment to a placebo experiment. In the placebo experiment, the households believe that they are in recession (they use their recession policy functions), but the underlying shocks hitting the economy are expansionary. More precisely, the aggregate state for employment risk is boom, securities and real estate prices are the same as during the 2002-2006 period, and there is no additional MIT labour income drop. The middle panel of Table 7 displays the model simulations for the baseline

	Cars ext. margin		Cars int. margin		Non-dur. cons.		Income	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Model vs. Data								
Model	-0.22	-0.42	-0.09	-0.19	-0.03	-0.05	-0.06	-0.10
Data	-0.21	-0.44	-0.12	-0.17	-0.02	0.00	-0.06	-0.10
Baseline vs. Placebo								
Baseline (model)	-0.22	-0.42	-0.09	-0.19	-0.03	-0.05	-0.06	-0.10
Placebo (model)	-0.10	-0.14	0.01	-0.02	0.00	0.00	0.0	0.01
Baseline vs. No Compo								
Baseline (model)	-0.22	-0.42	-0.09	-0.19	-0.03	-0.05	-0.06	-0.10
No compo. (model)	-0.21	-0.26	-0.09	-0.12	-0.03	-0.04	-0.06	-0.08

Notes: This table reports changes between boom (2002-2006) and the Great Recession (2008-2014) in the data and various model simulations. No compo. refers to the "no composition effect" experiment. Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Non-dur. cons. stands for non-durable consumption, ext. for extensive and int. for intensive. Income is income from labour and transfers. In the estimates from the data, top and bottom 1% are winsorised and households sampling weights are used.

Table 6: Great Recession Experiment

and the placebo experiments.

Unlike consumption and income, households' car purchases respond to the placebo scenario. Still, the nature of the response differs by type of contract. For permanent households, none of the baseline's cars intensive margin drop is accounted for by change in risk. Instead, the contraction in durables' intensive margin is only explained by realised income losses. Conversely, half of their baseline decline in the share of car buyers still takes place in the placebo experiment, implying that change in risk prompts permanent households to delay their durable purchases. Consequently, the change in permanent households' risk profile during the Great Recession shifts down their trigger adjustment point (s), without impacting their optimal durables stock (S). The story is different for fixed-term households as change in risk accounts for around a third of their baseline drop in cars extensive margin and for 10% of their drop in cars intensive margin. Change in risk shifts down fixed-term households' adjustment trigger (s) as well as their optimal durables stock (S).

The heterogeneity in the change of (S,s) rule's parameters across types of contract is explained by the heterogeneous nature and changes of the risk faced by the households. Table 7 displays moments of the income distribution in boom and during the

Great Recession by type of contract.¹²

	Mean		Variance		Skewness	
	Perm.	F.t.	Perm.	F.t.	Perm.	F.t.
Boom	1.035	0.544	0.011	0.049	-4.091	2.299
Recession	1.032	0.520	0.013	0.035	-4.402	2.430
Change	0.0	-0.04	0.24	-0.29	0.08	0.06

Notes: Perm. stands for permanent contract and F.t. for fixed-term or temporary contract. Boom is 2002-2006, Recession is 2008-2014. Income is income from labour and transfers.

Table 7: Moments of Households' Income Distribution

Permanent households and downside income risk

First, we consider permanent households. Consistent with Guvenen et al. (2021), the income distribution is negatively skewed. Upon entering the Great Recession, permanent households see that their average income remains the same, their income variance rises by nearly a quarter, and their income distribution becomes more negatively skewed. This increase in variance and negative skewness is driven by the rise in the probability to fall into unemployment or to become a fixed-term worker. There is an expansion of the lower tail of the income distribution as permanent households see a rise in the risk of large income declines. This risk pattern interacts with car expenses as the non-convex adjustment cost makes car purchases partially irreversible. In other words, buying a car is a commitment that permanent households will be less willing to make in an environment with large downside income risk. In such times, it is optimal for them to delay durable purchases and wait until the uncertainty subsides. Higher income uncertainty encourages permanent households to "wait-and-see", widening the inactivity region of their policy function. The rise in downside income risk is driving permanent households' car response over the Great Recession. The associated strategy is "wait-and-see".

Fixed-term households and upside income risk

Interestingly, Table 7 shows that fixed-term households' income distribution is positively skewed. This is the case as, for them, the main source of large income changes is to be upgraded to a permanent contract. Fixed-term households have a lot of room to move up. Upside income risk is an important feature of their income process. Upon entering the Great Recession, fixed-term households face a 4% decline

¹²These moments are computed without the MIT income drop.

in their income mean alongside a 30% drop in income variance. Moreover their income distribution becomes less positively skewed. This is the case as the probability to get upgraded to a permanent contract reduces significantly (from 10.4% to 6.3% during the Great Recession). There is a compression of the upper tail of the income distribution.¹³ In terms of durable consumption, the standard "wait-and-see" story is a poor description of fixed-term households' behaviour as they do not need to wait for the uncertainty to subside - it already has. Instead, households would like to downsize their stock of durables, but the non-convex adjustment cost makes selling cars extremely costly. Moreover because it is non-convex, the cost is particularly discouraging for low income low wealth households who are over represented amongst fixed-term contracts. As depreciation naturally reduces the value of cars over time, fixed-term households use this mechanism to downsize and save on transaction costs. In essence, they "wait-to-downgrade". The drop in upside income risk is driving fixed-term households' car response over the Great Recession. The associated strategy is "wait-to-downgrade".

Persistence of consumption contraction

Finally the driver of the drop in durable purchases - upside versus downside income risk - has a significant impact on the length of the contraction. Figure 3 plots changes in the extensive margin of car purchases for permanent households (in blue) and for fixed-term households (in yellow). The baseline period ($t=-1$ on the plot) is 2007, while the Great Recession starts in 2008 ($t=0$ on the plot) and lasts for the whole plotted period. On the one hand, for permanent households, the drop in the extensive margin of car purchases is relatively short-lived. As the recession draws longer, depreciation reduces the value of households' stock of cars. Eventually, this value sinks to such a low level that permanent households are forced into adjusting despite the ongoing recession. On the other hand, for fixed-term households, depreciation is a lengthy process. The "wait-to-downgrade" strategy is persistent and lasts throughout the recession.

¹³Fixed-term households also experience a small rise in unemployment risk during the recession. Still, the most important driving force is the decline in upside income risk as illustrated by the reduction in income variance and 90th percentile.

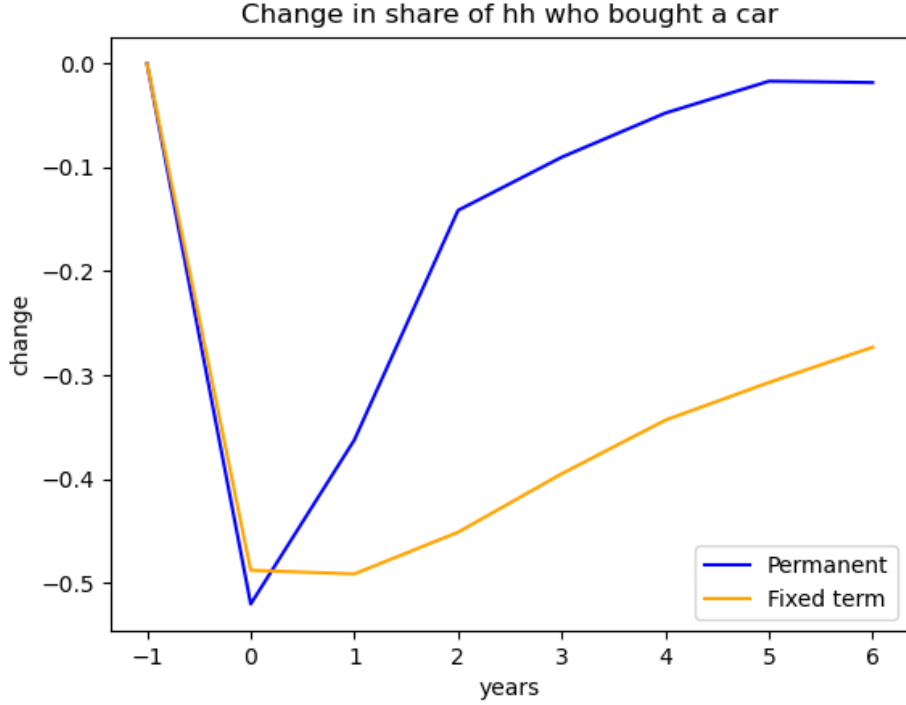


Figure 3: Persistence of "Wait-and-See" *versus* "Wait-to-Downgrade"

5.3 The Composition Effect

Beyond change in risk, households also adjust their saving and consumption patterns because they experience resources loss during the Great Recession. These losses come from two channels: i) the "exogenous losses" (wealth shocks and MIT income drop) and ii) the composition effect. The composition effect refers to which households are selected in the permanent and fixed-term groups. To be considered a permanent (fixed-term) worker, one needs to have been employed with a permanent (fixed-term) contract for at least 6 months in the year. Consequently, a share of households in the permanent (fixed-term) group will have gone through some spells of unemployment or fixed-term (permanent) contract in the period of interest. How many households do go through these alternative employment states depends on the Markov matrix governing employment transitions, which is state dependent.

To quantify the relative importance of the two resources loss channels, I compare the baseline experiment to a "no composition effect" experiment. In the "no com-

position effect" experiment, I use the employment state transition matrix calibrated for the expansion time, even during the Great Recession. Households still believe that their employment risk has changed (they use their recession policy functions) and the exogenous income and wealth losses are as in the baseline. This experiment removes the composition effect: the employment states of the households selected in a given group are exactly the same as if the Great Recession did not occur.

The bottom panel of Table 6 displays the results of the "no composition effect" experiment for permanent and fixed-term households. First, permanent households' baseline and "no composition effect" experiments are similar, implying that composition effect does not play a role for these households. Given the persistence of the permanent contracts, there is only a residual share of the selected households who have gone through any other employment state during the year. This is true in boom and in recession. Therefore, the sole change in the realised transition matrix does not affect the households selected in the permanent group much.

Conversely, the composition effect accounts for around 40% and 35% of fixed-term households' drop in car purchases extensive and intensive margins. It is the case as fixed-term contracts are significantly less persistent than permanent ones. Therefore, households selected in the fixed-term group are much more likely to have gone through other employment states over the period. In recession they are more likely to have gone through unemployment spells and significantly less likely to access a permanent contract.¹⁴ If the researcher were to omit the composition effect and attempt to match fixed-term contracts' car purchases extensive margin over the Great Recession, the car non-convex adjustment cost would be highly overestimated. For instance, one would need a non-convex adjustment cost of 35% (against 6.5% estimated in the paper) to obtain a close to 40% drop in fixed-term households' cars extensive margin. Such an adjustment cost is unreasonable.

Figure 4 provides an illustration of the nature and importance of the composition effect for fixed-term households by plotting the impulse response functions for the Great Recession baseline experiment (in blue), and for the "no composition effect" experiment (in orange).¹⁵ The baseline period ($t=-1$ on the plot) is 2007, while the Great Recession starts in 2008 ($t=0$ on the plot) and lasts for the whole plotted period. Fixed-term household's composition effect is subtle and evolves throughout the recession. The share of fixed-term contracts in the overall employed population is larger in recession than in boom. This implies that, in the first periods of the reces-

¹⁴The permanent contracts that are captured in the fixed-term group are often households who spent the start of the year with a fixed-term contract and upgraded to a permanent one after June. These upgrades are less likely in recession.

¹⁵Plots of permanent households' impulse response functions can be found in Appendix C.

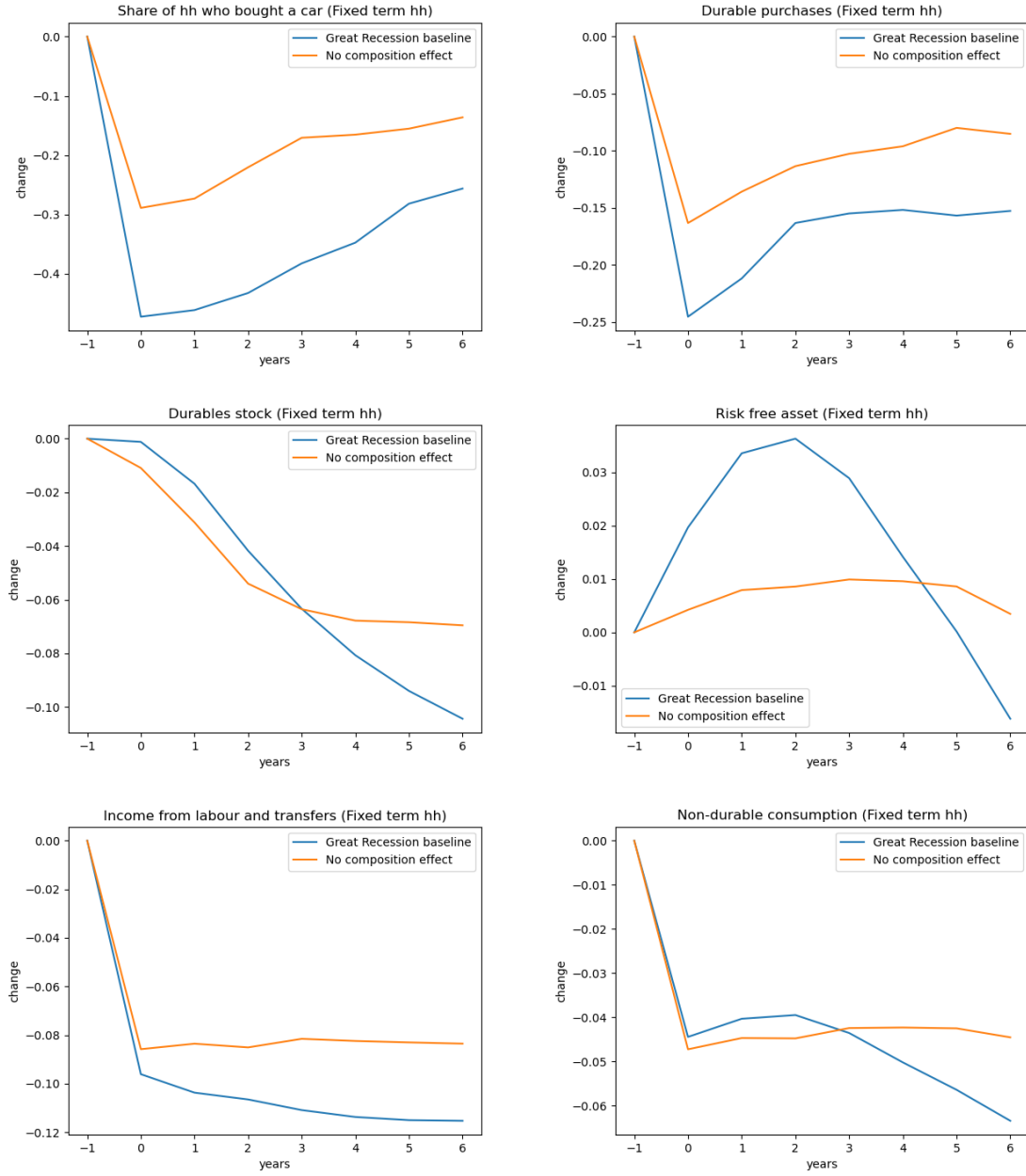


Figure 4: Impulse Response Functions (Fixed-term Households)

sion, the size of the fixed-term group grows (in the simulations, this is the case in the first two years of the recession). During this time, there is a flow of households tran-

sitioning from permanent contracts to fixed term contracts. On average, permanent workers have higher wealth and durables stock than their fixed-term counterparts. Consequently, at the start of the recession, the flow of permanent households into the fixed-term group brings up the group's average wealth and stock of cars. This composition effect can be seen in panels c and d of Figure 4 where average car stock and liquid wealth are larger in the baseline than in the experiment without composition effect. In the later part of the recession, when the size of the fixed-term group has stabilized, the change in its composition is only due to the larger share of unemployed households and (mostly) the lower share of permanent contracts.

Finally, the composition effect deepens the size of the car consumption drop along both adjustment margins. In the first part of the recession, the households who just transitioned from permanent contracts own a stock of cars that is too high compared to their current income and prospects. Consequently, they will not make new car purchases. For the later part of the recession and for the intensive margin, the lower share of permanent contracts in the fixed-term group reduces the adjustment frequency and the size of car purchases.

6 Policy Experiment: Car Purchase Subsidy

In 2009, the Italian government spent 2 billion euros in subsidies to support the automobile sector. To explore the consequences of such a policy, I use the model as a laboratory and investigate the impact a payment corresponding to 5% of car expenses during 2009. For the average car purchase, this corresponds to roughly €500. In the simulations, households did not anticipate the implementation of the subsidy. In 2009, they are aware of the programme, but they do not expect that it will last only for a year. It is important to note that, in this exercise, the programme is a "free lunch" as we are in partial equilibrium and there is no tax to finance the subsidy. Consequently, the results of the experiment cannot be directly compared to the reality. They provide an over-estimate of the positive impact of the policy.¹⁶ Still, this exercise is informative of how much car subsidies can achieve (at best), and of the potentially heterogeneous impact on households exposed to different type of risk.

The left panel of Figure 5 shows the share of households who bought a car in a given year for the permanent group (in blue) and for the fixed-term group (in yellow). The solid lines represent the simulations with the 2009 car subsidy, while the dashed lines are the Great Recession baseline experiment. First, in 2009, the subsidy

¹⁶If we assume that the aim of the policy is to stimulate car consumption.

strongly increases the share of buyers (44% for permanent and 27% for fixed-term households). However, as soon as the policy ends, there is a reversal and the share of buyers significantly drops for the following couple of years. The car subsidy did not trigger net new purchases, but simply prompted households to anticipate their car investment. The timing of purchases was brought forward, leading to depressed car consumption in the post-programme period. This finding is consistent with the empirical analyses of Mian and Sufi (2012), Hoekstra, Puller and West (2017), and Green et al. (2020), who study the cash-for-clunkers policy in the US and find that its stimulative effect on car consumption occurs mainly by changing the timing of investments rather than by inducing additional purchases.

What is more, the car subsidy even deepened the drop in the value of car expenses on the intensive margin. In the baseline experiment, cars intensive margin decreases by 9% and 19% for permanent and fixed-term households respectively. With the car subsidy these numbers jump to 14% and 24%. In 2009, the subsidy implies that a much larger share of households buy a car. Therefore, the average distance between the current car and the target one (i.e. the value of the purchase) is much smaller. After 2009, households' target stock of cars is well below what they currently own, and they shift resources towards liquid assets. In summary, the car subsidy brings forward car investment while depressing the extensive margin of car consumption in future periods and failing to stimulate the intensive margin.

Finally, the impact of the subsidy depends on the type of risk faced by the households. In 2009, the spike in fixed-term households' car purchases is milder than that of their permanent counterparts. Following the collapse in their upside income risk, fixed-term household adopt a "wait-to-downgrade" strategy. Their aim is to lower their car holdings and therefore, the subsidy has limited ammunition to make them upgrade. Moreover, the subsidy can even be seen as a *downgrading subsidy*. The right panel of Figure 5 shows the share of households who downgraded their car in the year (i.e. sold their car to purchase a cheaper one). In 2009, a share of fixed-term households take advantage of the subsidy to downgrade their stock of cars.

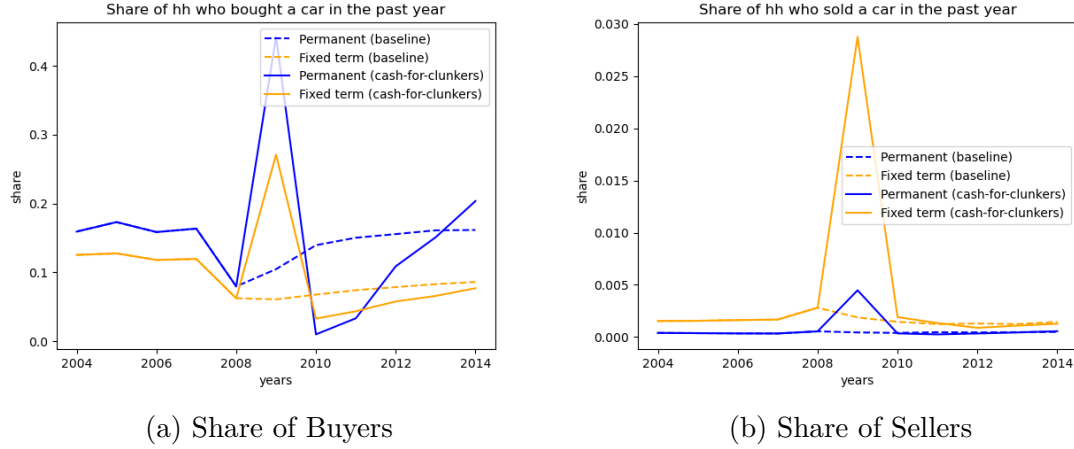


Figure 5: car purchase subsidy

7 Conclusion

This paper investigates the impact of risk heterogeneity on durable consumption over the business cycle. I examine household risk heterogeneity through the lens of employment contract type, and document that during the Great Recession, fixed-term contract workers reduced their car purchases significantly more than their permanent counterparts (along both the intensive and extensive margins). Using an incomplete market model with durable consumption and a dual labour market, I examine the drivers of the consumption patterns for each contract type. I find that, on the one hand, households that experienced an increase in downside income risk (such as permanent contract holders) adopted a "wait-and-see" strategy for their durable purchases during the Great Recession. On the other hand, households that experienced a decline in upside income risk (such as fixed-term contract holders) adopted a "wait-to-downgrade" approach to their durable spending. In addition, about 40% of the decline in fixed-term households' car consumption can be explained by a composition effect. Omitting this composition effect would lead to a significant overestimation of the car non-convex adjustment cost. Finally, a car purchase subsidy has a limited ability to stimulate durable consumption for households facing a decline in their upside income risk. The policy is *de facto* partly transformed into a *downgrading subsidy*. A direction for future research is to extend this framework to study the general equilibrium effects of sector-specific technological shocks on durable consumption.

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A Data Appendix

A.1 SHIW Data

This appendix provides additional discussion on the Survey on Households Income and Wealth (SHIW) conducted by the Bank of Italy between 1965 and 2020 (last available wave). I restrict my study to 2000-2014 because prior to 2000, the SHIW did not collect the data necessary for my analysis. From 2000, the SHIW provides detailed job-related characteristics, as well as information on households' income, wealth, non-durable consumption and durable goods expenditures. The SHIW also reports vehicles sales and purchases.

The survey is conducted every two years and has a panel component (that has been growing since its introduction and represents 55% of the sample in 2014). The units of observation are individuals and households. The sample size is approximately 8,000 households and 20,000 individuals in each wave. The survey provides sampling weights that are representative of Italian households and Istat computed monetary reevaluation coefficients.

The SHIW presents some advantages over American surveys that are often used to study durable consumption: the CEX and the PSID. The CEX has little information on households' characteristics and employment status, has a short panel element (households are part of the sample for a maximum of four consecutive quarters), and the frequency of income, wealth and consumption are not synchronised. On the other hand, the PSID also has a small sample size (2,000 households).

A.2 Sample Selection

Sample selection: Household-level by type of contract. I restrict my analysis to households with at least one employed income earner aged between 20 and 65 years old, without any self-employed member and without any currently unemployed member. Additionally, my analysis is restricted to households where employed members have the same type of contract or households with a single income earner. I exclude other households as they are difficult to sort in the permanent versus fixed-term contract groups. Finally I exclude households with missing information on type of contract. In Boom, the sample counts roughly 500 households in the fixed-term group and 7,500 observations in the permanent group. In recession, the fixed-term and permanent groups respectively gather 1,050 and 9,050 observations. This sampling is used for all the household-level variables by type of contract. However, I make an exception when I compute the intensive margin of car purchases for fixed-term households. To have enough observations, I keep any household where at least

one member holds a fixed-term contract, meaning that I keep households with two earners in different employment contract.

Sample selection: Household-level, all employment status. To compute population wide moments in the data, I restrict my analysis to households with at least one employed or unemployed member aged between 20 and 65 years old. I exclude household with any self-employed member.

Sample selection: individual-level. To study income and employment state transitions, I keep individuals currently employed or unemployed, aged between 20 and 65 years old, and who have been surveyed in at least two consecutive waves (three consecutive waves when I estimate the income risk process conditional on being employed). I exclude observations with missing information on the type of contract or missing income.

Sample selection: unemployed individuals. To recover unemployment benefit value and coverage, I select currently unemployed individuals who are aged between 20 and 65 years old. I exclude first-time job seekers as they do not qualify for unemployment benefits.

A.3 Variables Treatment and Definitions

To remove outliers, I winsorise the top and bottom 1% (by boom/recession period and by type of contract) for all numerical variables. For household-level variables, households sampling weights are used. Cars and non-durable expenditures are deflated using CPI price indices (CPI for new cars and CPI for non-durable consumption respectively). Other nominal variables are deflated using the Istat reevaluation index.

Definitions:

- **Car purchases (or stock)** encompass all means of transport including motorbikes and boats. In the data, the split between cars and other vehicles is available only from 2012 onward. For consistency, I keep cars and other vehicles as my measure of car purchases (or stock) throughout the entire period. This should not alter the results as between 2012 and 2014, cars represented more than 90% of all vehicles purchased by households. Car purchases (or stock) are normalised by dividing by OECD scale adult equivalent units.
- **Non-durable consumption** refers to all consumption expenditures excluding durable goods (cars, furniture, appliances). Non-durable consumption is

normalised by dividing by OECD scale adult equivalent units.

A.4 Summary Statistics

Table 8 provides some household-level summary statistics for 2002-2006 (before the Great Recession). Permanent households' average income and wealth is roughly twice as large as that of fixed-term households. Fixed-term households' heads are on average younger and less educated than the heads of permanent households. Finally, the share of fixed-term households is higher in the poorer regions of the South of Italy.

	Permanent contract	Fixed-term contract
Mean income (€)	38774	20613
Mean wealth (€)	188168	94264
Mean age of head of hh	44	40
Share living in the North (%)	53	32
Share living in the Center (%)	21	13
Share living in the South (%)	26	55
Share of college graduates (%)	11	6
Share of high school graduates (%)	46	26
Share living in a small city i.e. - 40,000 inhab. (%)	57	60
Mean number of children	1.23	1.33

Table 8: Summary Statistics

B Calibration

B.1 Employment State Transitions

Boom			
Target		Data	Model
Transitions (2-year time horizon)			
Perm. to Perm.		0.93	0.93
Perm. to F.T.		0.04	0.04
Perm to Unem.		0.03	0.03
F.T. to Perm.		0.49	0.49
F.T to F.T		0.36	0.36
F.T. to Unem.		0.15	0.15
Unem. to Perm.		0.23	0.23
Unem. to F.T.		0.15	0.15
Unem. to Unem.		0.62	0.62
Recession			
Target		Data	Model
Transitions (2-year time horizon)			
Perm. to Perm.		0.90	0.90
Perm. to F.T.		0.06	0.06
Perm to Unem.		0.05	0.05
F.T. to Perm.		0.33	0.33
F.T to F.T		0.47	0.47
F.T. to Unem.		0.20	0.20
Unem. to Perm.		0.14	0.14
Unem. to F.T.		0.13	0.13
Unem. to Unem.		0.73	0.73

Table 9: Targeted Moments of the Employment Transitions

Notes: Note that the Perm. to Perm. transition in the first column does not imply that the worker stayed in a permanent employment for two full years. It simply means that, at the two survey dates, the worker was employed under a permanent contract (he could have gone through other employment states in-between the two surveys).

B.2 Calibration of the Income Risk

I use the following restrictions to identify the persistent component's intercept, persistence and variance parameters:

$$\begin{aligned}\frac{Cov(\tilde{y}_{it}, y_{it-2})}{Cov(\tilde{y}_{it}, y_{it-1})} &= \tilde{\rho} \\ Cov(\tilde{y}_{it}, y_{it-1}) &= \tilde{\rho} * \sigma_P^2 \\ (1 - r\tilde{\rho}^2) * \sigma_P^2 &= \sigma_u^2 \\ E(\tilde{y}_{it}) &= \frac{\tilde{\mu}}{1 - \tilde{\rho}}\end{aligned}$$

The normalised income grids for each type of contract are displayed below.

Permanent contract:

$$\text{Income} = \begin{matrix} y1 & y2 & y3 & y4 & y5 \\ (0.55 & 0.74 & 1 & 1.34 & 1.81) \end{matrix}$$

Temporary/Fixed-term contract:

$$\text{Income} = \begin{matrix} y1 & y2 & y3 & y4 & y5 \\ (0.27 & 0.36 & 0.48 & 0.65 & 0.87) \end{matrix}$$

C IRFs (Permanent Households)

Figure 6 displays permanent households' impulse response functions for the Great Recession baseline experiment (in blue), and for the "no composition effect" experiment (in orange).

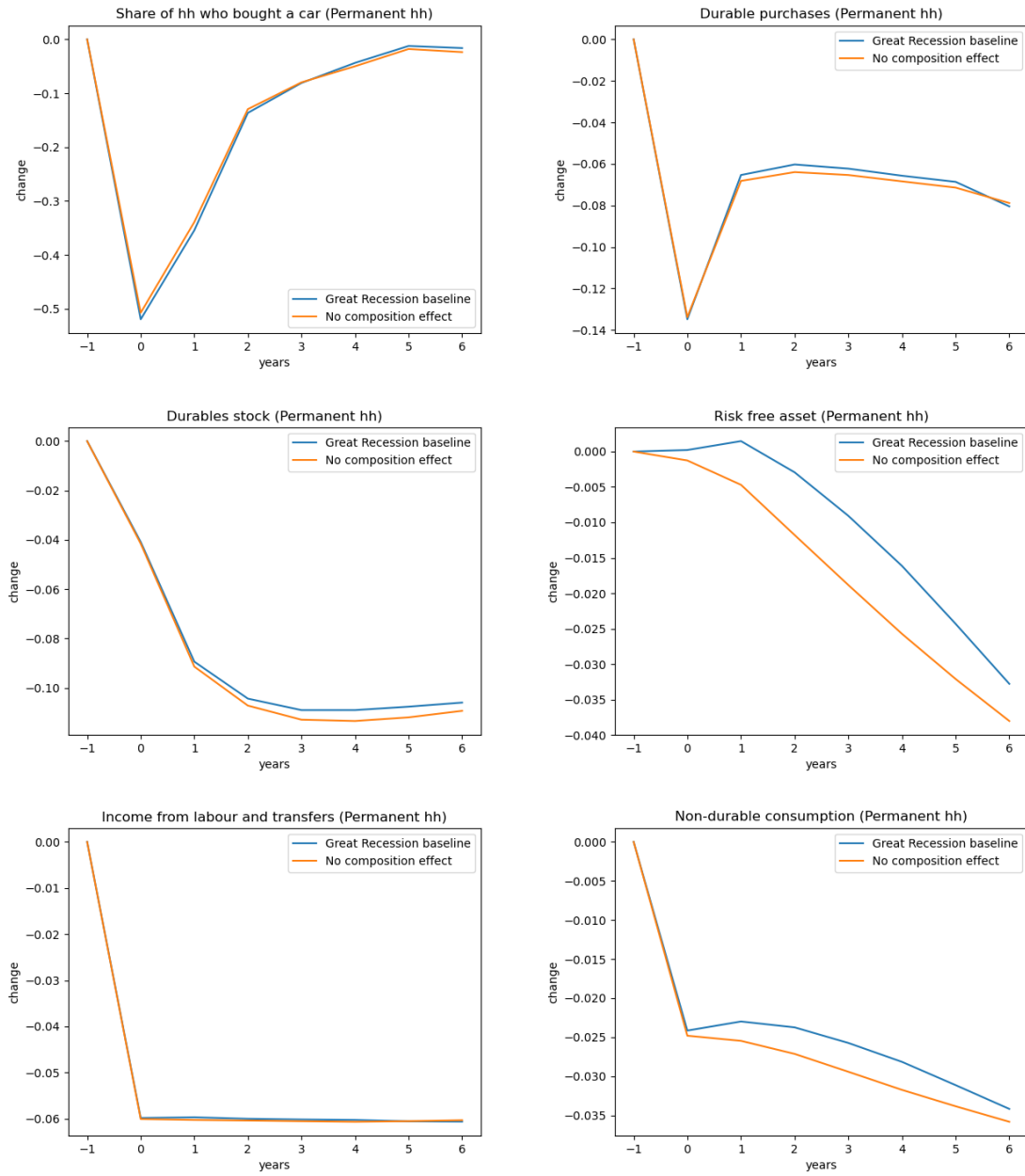


Figure 6: Impulse Response Functions (Permanent Households)